Fiscal Year:	FY 2019	**Task Last Updated:**	FY 11/20/2018
PI Name:	Burma, Sandeep Ph.D.		
Project Title:	Mechanistic Analysis of Particle Radiation-Induced Carcinogenesis Using Validated Mouse Glioma Models		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: Formerly at University of Texas Southwestern Medical Center at Dallas until fall 2019.		
Project Type:	Flight	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-RADIATION. Appendix D: Ground-Based Studies in Space Radiobiology
Start Date:	01/20/2016	**End Date:**	08/06/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):			
Grant/Contract No.:	NNX16AD78G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Galactic Cosmic Rays (GCR) and Solar Particle Events (SPEs) are major sources of carcinogenic risk for astronauts on prolonged space missions. Large uncertainties exist in the exact estimation of cancer risks from charged particles due to the paucity of epidemiological data in this area. Furthermore, a mechanistic understanding of genetic changes underlying transformation by particle radiation is necessary for the development of appropriate countermeasures. Thus, experimental animal models that closely mimic the process of carcinogenesis in humans are essential for oncogenic risk assessment. In the course of our previously funded 3 yr. NASA project (2013–2016), we characterized two sensitive and complementary mouse models that can be used to elucidate molecular mechanisms underlying the process of particle radiation-induced glioblastoma (GBM) development. GBMs are lethal brain tumors with very dismal prognosis for which radiation is the only known risk factor. GBMs represent the third leading cause of cancer-related death among adults aged 30 – 50 years (the average age of astronauts), the mean survival after diagnosis being only about 14 months. GBM was the first cancer to be analyzed by The Cancer Genome Atlas Network (TCGA), and the key genetic alterations occurring in GBM are now well defined. Based on this information, we utilized transgenic mice with brain-restricted deletions of GBM-relevant tumor suppressors in logical combinations to analyze the process of charged particle-induced carcinogenesis. We identified and characterized two complementary mouse GBM models that would be ideal for studying particle radiation-induced carcinogenesis -- Nestin-Cre; Ink4ab-/-Arf f/f, and Nestin-Cre; p53f/+Ptenf/+. These models exhibit a low frequency of spontaneous GBMs but readily develop brain tumors after exposure to HZE (high energy) particles, which resemble human high grade gliomas in their genetic and molecular signatures. Using these mice, we have clearly demonstrated that high-linear energy transfer (LET) charged particles have a greater transforming potential compared to low-LET radiation, and have characterized the molecular events underlying Fe ion-induced gliomagenesis in these models. This work sets the stage for quantitative studies on radiation quality effects on carcinogenesis for an array of charged particles representing the GCR (as proposed in Aim 1). These charged particles induce complex DNA double-strand breaks (DSBs) the accurate repair of which is critical for preventing transformation. Our models will also allow us to determine which DSB repair pathway represents a major barrier to charged particle-induced gliomagenesis (as proposed in Aim 2). This information would be fundamental to the development of a mechanistic understanding of charged particle-induced carcinogenesis. We will cross the transgenic mice characterized by us with mice harboring brain-specific ablation of non-homologous end joining (NHEJ) or homologous recombination (HR) repair pathways -- Nestin-Cre; Ligase4f/f, and Nestin-Cre; Brca2f/f – that were established in the laboratory of Prof. Peter McKinnon (collaborator on project). Finally, genomic and bioinformatics analyses will be used to define genetic signatures unique to particle-radiation induced glioblastomas and to delineate key pathways that are de-regulated during HZE-induced gliomagenesis (as proposed in Aim 3). We are hopeful that an in-depth mechanistic analysis of charged particle-induced gliomagenesis in these validated mouse models will help us to make significant progress towards cancer risk assessments and development of protective strategies for charged particle exposure. ED. NOTE August 2020: Principal Investigator Dr. Sandeep Burma moved to University of Texas Health San Antonio in Fall 2019 and Grant continues there
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	As cancer therapy with particles (protons and C ions) is becoming more widely used, our research with heavy ions is also important from a therapeutic standpoint here on Earth. Our examination of the relationship between LET and gliomagenesis could have practical implications for radiotherapy modality choice.
Task Progress:	The most comprehensive data set correlating linear energy transfer (LET) with carcinogenic potential comes from the mouse Harderian gland studies carried out in the 1980s. These studies indicate an initial linear relative biological effectiveness (RBE)-LET relationship that seems to plateau at higher LETs or doses. These and other in vitro studies indicate a maximal LET for cellular transformation at or around 100 keV/µm. As humans do not have Harderian glands, it is important to also use mouse models of human-relevant cancers to quantitatively assess radiation quality effects on carcinogenesis. The two complementary mouse models that we have characterized in the course of our previous 3-year NASA project (2013-2016) -- Nestin-Cre; Ink4ab-/-Arf f/f, and Nestin-Cre; p53f/+Ptenf/+ -- are uniquely suited to further elucidate the role of radiation quality effects on charged particle-induced carcinogenesis. These models exhibit a low frequency of spontaneous glioblastomas (GBMs) but are susceptible to radiation-induced brain tumors which resemble human high grade gliomas histopathologically and molecularly. Indeed, exploratory studies with four different ion species (H, C, Si, and Fe) clearly show that charged particles possess a greater gliomagenic potential relative to low LET X-rays. As a continuation of this work, we are evaluating the relative contributions of LET and atomic number (Z) to carcinogenesis and obtaining specific RBE values for representative charged particles using our two mouse models. Previously, we found that irradiation with Si (Z=14; LET=79 keV/µm) resulted in the highest incidence of radiogenic gliomas while irradiation with heavier Fe ions (Z=26; LET=174 keV/µm) resulted in lower tumor prevalence. These results indicate that Si (or LETs of ~79 keV/µm) is the most detrimental with respect to carcinogenesis out of all the ions we tested. However, there is a substantial gap in the Z and LET values between Si and the other species (C and Fe) that we tested. Thus, additional species with Z and LET values that fall in between C and Si and between Si and Fe were selected to: a) confirm the initial linear response between LET and tumor frequency, and b) to establish that Si with LET of ~79 keV/µm is one of the most detrimental ions in terms of cancer risk. To fill-in this gap and substantiate our existing results we chose two additional ions: Ne (Z=10; LET=39 keV/µm) and Ti (Z=22; LET=120 keV/µm). During the NASA Space Radiation Laboratory (NSRL) 16B campaign (first campaign funded by NNX16AD78G) Nestin-Cre; Ink4ab-/-Arf f/f mice were irradiated with 0.5 Gy of Ne ions (~40 mice) or Ti ions (~20 mice). Additional 20 animals were irradiated with Ti during NSRL 17A to complete the cohort. Following IR, the mice were shipped back to UTSW (University of Texas Southwestern Medical Center) where they are monitored daily for neurologic symptoms of gliomagenesis until the end of the study (9 month post IR). The Ne study ended in March 2017. Tumor frequencies obtained with Ne confirm the linear relationship between LET and tumor frequency and affirm that Si is the most gliomagenic of the ions investigated thus far. Results from mice irradiated with Ti during NSRL 16B and 17A clearly place Ti between Si and Fe. These results confirm Si as having the most gliomagenic potential concluding this aim of the study. Please note that all the ions tested thus far trigger high grade gliomas and show differences only in tumor frequencies. In addition, in order to understand the effects of LET versus atomic number (Z) on tumor frequency, we irradiated multiple cohorts of Nestin-Cre; Ink4ab-/-Arf f/f mice with Si ions of different energies and LETs during the NSRL 16C campaign. In addition, in order to obtain more accurate RBE values for Si, we irradiated cohorts of Nestin-Cre; Ink4ab-/-Arf f/f mice with a range of doses of Si ions (79 keV/µm) and a range of energies and LETs during the 2017 campaigns. The results show that a single low dose of 0.1 Gy

was sufficient to trigger GBM development in this mouse model and that tumor frequencies increased with increasing dose thereafter. Additionally, we found that GBM frequencies increased with increasing LET (decreasing energy), illustrating how changes in the LET of the same ion species (Silicon) can affect tumor development.

We are currently using the simplified GCR (H, C, O, Si, and Fe) simulations to more faithfully elucidate the effects of space radiation on GBM development in the Nestin-Cre; Ink4ab -/-, Arf f/f and Nestin-Cre; p53f/+Ptenf/+ models. Mice are being monitored for tumor formation. Moreover, we are also using our mouse models to ask which DNA repair pathway is critical for preventing particle radiation-induced gliomagenesis. Towards this end, we are crossing two inducible Cre models, GFAP-Cre-ERT2;p53f/f and PlazG7-Cre-ERT2;p53f/f transgenic mice, with mice harboring brain-specific ablation of NHEJ or HR repair pathways -- Ligase4f/f and Brca2f/f – developed in the laboratory of Prof. Peter McKinnon of St. Jude's Children's Research Hospital (collaborator on project). The initial breeding has been established in 2017-18, and mice colonies are now being expanded for irradiation in NSRL 2019 campaigns.

We have also carried out preliminary whole genome sequencing studies to determine whether radiation quality could contribute to distinct mutational signatures in gliomas derived from different LETs. We found that genome-wide there was a qualitative difference in the scale of deletions between electromagnetic and particulate radiation-induced GBMs. More extensive sequencing efforts are planned for 2019. Finally, we are focusing our genetic signature studies to include RNA and microRNA changes in the brain and plasma induced by charged particle radiation. We hypothesize that identification of such RNA and microRNA signatures could be used as early biomarkers for charged particle radiation exposure and for early detection of radiation-induced gliomagenesis. Also, these analyses will provide mechanistic insights into how changes in the brain microenvironment might promote tumor development and help explain the pre-ponderance of certain genetic changes (like Met amplification) in the radiogenic gliomas.

ED. NOTE August 2020: Principal Investigator Dr. Sandeep Burma moved to University of Texas Health San Antonio in Fall 2019 and Grant continues there. See that project for subsequent reporting.

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